Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/16/2025
PI Name:	Gordon, Gwyneth Ph.D.		
Project Title:	Evaluating Resistive Exercise as a Long-term Countermeasure for Spaceflight-induced Bone Loss Using Calcium Isotopes		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Tempe	**State:**	AZ
Zip Code:	85287-1404	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	04/11/2022	**End Date:**	04/10/2035
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	The previous Principal Investigator (Stephen Romaniello, Ph.D.) left the project in September 2021. The new PI is Gwyneth Gordon, Ph.D., based at the School of Earth & Space Exploration, Arizona State University. The investigation continues with the same project title as before, but a new grant number has been assigned Grant #80NSSC22K0956. See Grant #80NSSC19K0519 for prior project information under Dr. Romaniello (Ed., 9/12/25).		
COI Name (Institution):	Anbar, Ariel Ph.D. (Arizona State University) Smith, Scott Ph.D. (NASA Johnson Space Center) Zwart, Sara Ph.D. (KBR Wyle / NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC22K0956		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

We have previously demonstrated that variations in naturally-occurring stable isotopes of calcium (Ca) in blood and urine can be used to monitor changes in net bone mineral balance (BMB) during bed rest and spaceflight, including crewmembers using resistive exercise and pharmaceutical interventions to prevent bone loss aboard the International Space Station (ISS). Here, we propose to build on our established dataset of Ca isotope variations and BMB in International Space Station (ISS) crewmembers to include future crewmembers participating in the One-Year Mission Project. This proposal builds on a successful existing collaboration between researchers at Arizona State University (ASU) and NASA Johnson Space Center (JSC) to study and apply the Ca isotope method as a bone biomarker. [Ed. Note: "The One-Year Mission Project" was the previous name for "CIPHER". "CIPHER" is the short title for a set of 14 studies sponsored by NASA and international partner agencies (https://). CIPHER stands for "Complement of Integrated Protocols for Human Exploration Research".]

The primary aim of this project is measure Ca isotope data from urine and blood samples to be obtained in the proposed project; this data will characterize the trajectory of changes in crewmember BMB resulting from 2-12 months of spaceflight aboard the ISS and will provide a basis for extrapolation of expected changes in BMB and whole-body bone mineral content during missions extending 2-3 years in duration. Existing data from 6-month ISS missions suggest that crewmembers using the Advanced Resistive Exercise Device (ARED) experience rapid net bone loss during the first 60 days of flight, but that the rate of bone loss slows during flight days 60-180. If this trend continues, little additional cumulative bone loss might be expected during extended missions, consistent with the principle of non-inferiority. Ca isotopes are uniquely suited to answer this question, since they are the only technique currently capable of providing a time series of quantitative estimates of net BMB from samples returned from a single spaceflight mission.

The secondary aim of the project is to better constrain the statistical distribution of bone loss in ISS crewmembers. Limited existing data suggest that the observed distribution is skewed toward rare individuals who experience unusually high amounts of bone loss. Accurately evaluating risks to musculoskeletal health requires that NASA consider this high bone loss tail.

The third aim of this project is to coordinate, measure, and distribute data for all the biochemical measurements of the CIPHER (Complement of Integrated Protocols for Human Exploration Research) program. It is only through aligning samples and data sharing across many studies that we can holistically understand the range and magnitude of human biological responses to microgravity of varying duration.

Our proposal falls under the "Musculoskeletal system" and "Food and nutrition" Human Research Program (HRP) research emphases.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Project entered into the NASA Task Book in 2025 (Ed., 9/12/25).
The CIPHER complement of studies was envisioned to be the most ambitious, holistic evaluation of astronaut health ever conducted. The major focus has been on risk assessment for long-term spaceflight in support of successful expeditions for crewed missions to Mars. The original vision remains, but the study has had to pivot to accommodate financial, cultural, and logistical challenges to the NASA mission.

The proposed set of original studies was highly ambitious in terms of the breadth of studies, the number of astronauts, and the comprehensiveness of the data to be collected. Due to reductions in the number of astronauts per crew, the earlier implementation of commercial spaceflight missions transporting crew (and the incompatibility of custom spacesuits across commercial platforms), the CIPHER program has had to undergo several major rescopes. The first of these was in February 2022, when feedback from NASA’s crew office and logistical challenges required descoping the program and reducing the number of studies.

To maximize the science from the reduced number of studies, several research directions and groups were added to this project for organizational purposes in the process of this rescope. Drs. Liphardt and Niehoff of the German Sport University of Cologne and Dr. Kuro-o of the University of Tokyo were added to the Institutional Research Board of PI Gordon’s study. Drs. Liphardt and Niehoff's research, supported by the German Aerospace Center (DLR), was evaluating cartilage damage through collagen biomarkers during spaceflight, which can have serious implications for mission success for astronauts, particularly because cartilage turnover and repair are substantially slower than other tissues. Dr. Kuro-o and his team, financially supported by JAXA, planned to look at L-FABP, a biomarker of renal injury. Bone resorption during microgravity and limited mechanical loading can flood the bloodstream with excess calcium and phosphorus, placing a strain and potentially compromising kidney function. These additions required major modifications to all the documents of the Institutional Research Board (IRB) approval of the study, essential for ensuring ethical human research. Scripted videos of each study in the CIPHER complement ensured that the information provided to astronauts was both accurate and presented in a noncoercive manner. Briefings were presented to astronaut crews, and we began to consent astronauts to participate.

Later, Dr. Kuro-o and his team decided to withdraw the JAXA study from the CIPHER complement. This required additional modifications to the IRB to remove their research from the study. These modifications were approved in September 2025.

In parallel, we have been working on aligning data and sample sharing between the many different projects to maximize the science that can be conducted while minimizing the impact on busy astronauts. This includes planning on exact volumes of blood draw, and specifics of how many microliters each study is provided. In addition, because many of the research areas are complementary, it means that often the same data is required by multiple studies. For each component, such as ionized calcium concentration, lipid profiles, or biomarkers of bone formation and resorption, one group is designated to complete the measurements. However, very detailed plans about which other studies must receive the data while protecting the personal information from the astronauts must be negotiated.

The Arizona State University laboratory had planned to complete the measurements of calcium isotopes in blood and urine using our validated and established measurement modality of MC-ICP-MS after sample purification. Unfortunately, our instrumentation was nearly 20 years old, far beyond the anticipated lifetime of 8-10 years. In addition, the manufacturer stopped making the instruments, meaning that spare parts, repairs, and technical support became extremely difficult to obtain, and we realized that the instrument could not be reasonably expected to perform at

the high levels needed for these difficult measurements. An augmentation grant to cover a portion of the purchase of a replacement instrument (Thermo Fisher Neoma MC-ICP-MS) was provided, with more than 60% of the purchase price coming from multiple other charitable, institutional, and federal sources.

Suitable laboratory renovations were overseen by Principal Specialist Tyler Goepfert, and the instrument was delivered in December 2023 with installation in early 2024. This replacement instrument was a substantial change from previous models, and ASU staff worked with the manufacturer to ensure acceptable performance and workflow. Over the last 8 months, we have been validating the instrument's performance of calcium isotopes to capture the high-precision and accuracy data required. In addition, from independent funding sources, we were able to obtain a new microFAST Isotope 2 sample introduction system. This doubles sample throughput with a syringe pump system and has only recently reached the market. PI Gordon presented some of the initial validation of the improved sample throughput, sensitivity, and quality control metrics at two separate talks at the Goldschmidt conference in July 2025. Again, on separate funding, she will present additional validation at the Winter Conference on ICPMS in January 2026.

Samples are to be delivered in batches. No biological samples have yet been delivered to ASU, so efforts have been focused on improving workflow and data precision so that there will be seamless data delivery when samples begin to arrive. To date, seven astronauts have consented to participate in the CIPHER complement of studies. In summer 2025, NASA reevaluated the organization's scientific priorities, pivoting away from risk assessment towards risk management during long-term spaceflight. This has led to the difficult decision to continue the data collection with the existing number of astronaut participants, but to discontinue consenting new astronauts into CIPHER. In addition, there have been some reductions in the data collected, but all study PIs are working hard to capture the maximum high-quality scientific results despite these constraints. The original vision of ten astronauts each in 3-month, 6-month, and 12-month International Space Station (ISS) missions will not be met, but we anticipate that this will still produce incredible new knowledge into the impact of long-term spaceflight on human health and prepare humanity to extend crewed missions far into the solar system. Although the original vision of CIPHER may not come to fruition, it will still be the largest, most holistic study of astronaut health in response to long-term spaceflight ever conducted.

Bibliography Type:	Description: (Last Updated: 10/02/2025)
Conference Materials (Downloadable)	Gordon GW, Romaniello SJ, Smith SM, Zwart SR, Letcher AJ, Anbar AD. "Bone health: An integrated approach to monitoring the dynamic net bone mineral balance during 2-, 6- and 12-month missions aboard the ISS." 2022 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-10, 2022. , Feb-2022 Gordon_IWS2022_BoneHealth.pdf
Conference Materials (Downloadable)	Gordon GW, Romaniello SJ, Smith SM, Zwart SR, Letcher AJ, Anbar AD. "Bone health: an integrated approach to monitoring the dynamic net bone mineral balance during 2-, 6- and 12-month missions aboard the ISS." 2022 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-10, 2022. , Feb-2022 Gordon_IWS2022_BoneHealth_movie.mp4
Conference Materials (Downloadable)	Gordon GW, Smith SM, Zwart SR, Letcher AJ, Anbar AD. "Bone health: An integrated approach to monitoring the dynamic net bone mineral balance during 2-, 6- and 12-month missions aboard the ISS." Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023 Gordon_IWS2023_BoneHealth Final.pdf
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Conference Materials (Downloadable)	Gordon GW, Goepfert TJ, Manhart MW, Field PM "Isotope 2: Precision, stability and throughput improvement in Ca isotopes" Abstracts. 2025 Goldschmidt conference, Prague, Czech Republic, July 6-11, 2025. , Jul-2025 Goldschmidt 2025_Isotope2.pdf
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